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TMENet: Multimodal Road Crack Segmentation Network for Complex Scenarios

Ruoli Yang, Jie Jiang, Lixin Zhan, Baiyan Wan, Rui Qi, and Guoxiong Zhou

***Abstract*—Road crack detection in complex scenarios is challenged by road surface backgrounds such as water stains, oil stains, and printed road signs. To better address these issues, we propose a Multimodal Road Crack Segmentation Network–TMENet, based on complex scenes. In this article, we introduced text guidance in road crack segmentation tasks to supplement more detailed crack features. Firstly, we designed the Textual Crack Enhancer module(TCE Module) to fine-tune the Text Transformer in the Clipseg model to enhance its ability to parse specific crack descriptions. Secondly, we designed a multiscale Mamba module(Ms-Mamba Module) that integrates multiscale composite convolution and residual connections to improve the sensitivity of the model to local and multiscale details and enhance the network's ability to recover lost features. Then, we designed a Laplacian edge detection module (LED module) that combines deep visual features and high-frequency edge information to enhance the crack edge detection capability in complex scenes through a refined feature adjustment mechanism. Finally, TMENet achieved the best results in comparative testing with seven state-of-the-art segmentation networks on the CampCrack800, AsphaltCrack300, Crackseg9K, and CRKWH100 datasets. In addition, TMENet has achieved excellent results in actual road testing, and the test results show that TMENet is an effective auxiliary method to improve the accuracy of road safety detection.**

***Index Terms*—Complex scenes, road crack segmentation, text guidance, Clipseg, multimodal, fine-tune.**

I. Introduction

With the maturity of road construction technology, the scale of transportation road networks continues to expand, and regular testing of road safety and durability is crucial. Road cracks can lead to a decrease in structural load-bearing capacity. Therefore, road crack detection has become one of the most fundamental research topics in road maintenance [1], [2], [3], [4].

With the advancement of computer vision technology, image-based non-destructive testing (NDT) has become a widely used method for road crack detection. Traditional image processing methods, such as edge detection and threshold segmentation, have been explored for automatic crack recognition. However, these methods often struggle to cope with complex road conditions and lighting changes, making deep learning based methods more promising.

In recent crack segmentation tasks, due to the irregularity of crack morphology and complex background interference, attention distribution is often uneven, resulting in the model forgetting or misjudging some features. Qu et al. [70] proposed MFDB Net, which achieves multi-scale fusion of crack features through a multi-attention dual-branch architecture, demonstrating strong robustness under complex background conditions. Qu et al. [71] developed a directional connectivity feature enhancement network that effectively captures the connectivity patterns of cracks, thereby improving the continuity and accuracy of crack detection. To address these issues, researchers have proposed a series of attention optimization strategies. Sun et al. [42] proposed a Multiscale Attention Module based on DeepLabv3+, which dynamically adjusts the weight allocation of high-level and low-level feature maps by generating Attention Masks to reduce feature loss. Gao et al. [43] proposed the Atrous Spatial Pyramid Fusion Attention Module (ASAM) to adaptively allocate channel weights from a multiscale perspective, taking into account contextual semantic information, in response to the complexity of the multiscale features of cracks. Qu et al. [34] used the Global Context block (GC) to model the global information of cracks and combined it with multiscale feature extraction to improve detection performance. Liu et al. [44] proposed CrackFormer-II, which is an optimized version based on Self-attention and introduces a Fusion Module for Scaled Attention to suppress non-semantic information. Wang et al. [46] proposed the Residual Efficient Attention Structure as an alternative to traditional convolutional encoding blocks. In the encoding stage, this structure can adaptively adjust the attention weights of different regions.

However, existing attention still faces the challenge of achieving an effective balance between global and local features. On the one hand, modeling global semantic information enhances the overall perception ability of the model, but in complex backgrounds, this globality leads to the masking of edge and detail information of cracks; However, when the locally enhanced attention mechanism captures local features, it often lacks global information guidance, resulting in local features being easily lost due to background noise interference.

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Ruoli Yang, Jie Jiang, Lixin Zhan, Baiyan Wan, and Rui Qi are with the College of Systems Engineering, National University of Defense Technology, Changsha 410073, China (e-mail: yrl_14@nudt.edu.cn; jiejiang@nudt.edu.cn; zhanlixin98@outlook.com; wanbaiyanone@nudt.edu.cn; qirui20@nudt.edu.cn).

Guoxiong Zhou is with the College of Computer and Information Engineering, Central South University of Forestry and Technology, Changsha 410004, China (e-mail: zhougx01@163.com).

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In order to accurately extract crack edge information and suppress background noise in complex backgrounds, Lin et al. [47] proposed MLP Tagging, which projects high-dimensional crack features into a low-dimensional space to reduce redundant information. Dong et al. [50] designed a spatial convolutional neural network, Spatial CNN, to enhance the ability to extract spatial relationship features in complex backgrounds. Zheng et al. [51] proposed the TV NetTensor Voting Network, which extracts unique structural features of cracks by generating feature maps while reducing background interference. Liu et al. [49] proposed a Hierarchical Network Architecture based on ResNet and introduced MultiScale Fusion at the network head to ensure that crack edge details are not lost. In addition to the above methods, some researchers have explored the combination of morphological methods and deep learning to utilize the advantages of both to optimize crack detection fully. Yang et al. [52] proposed a crack detection method based on Fully Convolutional Network (FCN) and optimized the crack detection results by combining morphological operations. Sun et al. [53] proposed a CCSNet based on Weber's Law. CCSNet cleverly solves the problem of uneven grayscale values in road surface images after rain by using a grayscale directional adjustment algorithm. In the above methods, morphological information is used to supplement and enhance the expression of crack edge features, improve the detection ability of small cracks, and, to some extent, reduce the interference of background noise. However, the above method independently applies morphological filtering in the post-processing stage, and this offline processing method cannot dynamically fuse morphological information in the feature extraction process.

When segmenting road cracks, there are still three issues that urgently need to be addressed: (1) In a single crack image segmentation task, the image itself may not provide sufficient contextual information to understand the characteristics of the cracks, such as the severity of the cracks. Therefore, the features of a single image modality cannot fully represent the characteristics of the segmentation target. (2) The network exhibits different levels of attention when processing different image information, and due to the uneven distribution of attention, it is easy to miss some key information. (3) In complex scenarios, networks are easily influenced by background information. Therefore, it is difficult to accurately grasp the differences in edge features of cracks in complex scenes.

To address these issues, this paper proposes a multimodal road crack segmentation network for complex scenarios (TMENet). By integrating text and image features, dynamically allocating attention in different regions, and enhancing edge detection, this method aims to achieve more robust and accurate road crack segmentation. The contributions of this article can be summarized as follows:

(1) To solve the problem that the crack features extracted from image modalities cannot fully represent all the features of cracks, we first used a text image multimodal network in road crack segmentation tasks to introduce text guidance to supplement more detailed crack feature information. We designed a textual crack enhancer module(TCE Module) to fine-tune the Text Transformer in the Clipseg model to enhance its analytical ability for describing specific cracks.

(2) To solve the problem of crucial local information being relatively forgotten in the crack segmentation task of Vision Transformer due to the imbalance of global and local information. We designed the multiscale mamba module(Ms-Mamba Module) to capture and compensate for local features at different scales comprehensively. The Ms-Mamba Module improves the sensitivity and restoration ability of the model to local and multiscale details in segmenting crack images by integrating multi scale composite convolution and residual connections.

(3) To improve the poor detection performance of road crack edges in complex backgrounds, we designed a Laplacian edge detection module (LED module) to enhance the crack edge detection capability in complex scenes. This module integrates Laplacian morphological edge information into deep vision features. It improves the ability to distinguish crack edges from background noise through a refined feature adjustment mechanism, thereby enhancing the robustness and accuracy of crack detection.

(4) We have carried out a lot of experiments to verify the effectiveness and progressiveness of the network. Firstly, we investigated the impact of different text lengths on crack segmentation results. Analyzed the crack features contained in the text that can provide optimal guidance for crack segmentation. Secondly, we conducted ablation experiments on TMENet to verify the effectiveness of each module. Then, we verify the progressiveness of TMENet by comparing it with the seven most advanced partition networks. Then, we conducted generalization experiments on four other types of publicly available crack datasets. Finally, we tested TMENet in actual road environments. The experimental results demonstrate that it can accurately identify and segment road cracks in complex scenes.

The structure of the subsequent sections of this article is as follows. Section II is about the related research work on pavement crack segmentation. Section III introduces our self-constructed dataset, CampCrack800, and provides an overview of the TMENet, including its overall architecture and fundamental components. Section IV presents all the datasets used in our study and analyzes the experimental results in detail. Section V discusses the limitations of our method and its potential impact on intelligent transportation systems. Finally, Section VI concludes the article, summarizing key findings and insights.

II. Related Work

A. Segmentation Method Based on Traditional Image Processing

With the development of computer vision technology, image-based non-destructive testing has become a research hotspot for road crack detection. Traditional methods such as Canny operator, Sobel operator, edge detection, threshold segmentation method, etc. [5], [6], [7], and [8] can compensate

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for the strong subjectivity and high labor intensity of traditional manual detection. However, these methods require manual feature design and have poor generalization [8]. Due to the strong detection of road edges in traditional methods, there is less background noise [9]. Therefore, traditional crack edge detection algorithms perform better for these crack images. However, advanced image analysis techniques have been proposed for more complex scenarios such as asphalt or road surfaces with a large amount of unclean backgrounds [10], [11], [12], [13], [14]. However, these methods were highly dependent on manually designed features, making them less adaptable to varying crack appearances, lighting conditions, and complex backgrounds. In addition, Qu et al. [72] proposed a tunnel crack image processing method that combines specific statistical pixel ratios with adaptive segmentation. This approach effectively suppresses spurious linear crack patterns in tunnel images and enhances the overall integrity of crack representation.

B. Crack Segmentation Method in Deep Learning

Convolutional neural network (CNN), as one of the earliest and widely used structures in deep learning, has been extensively studied in the early stages due to its superior image feature extraction ability. Through structural improvement and module fusion, the performance of crack detection has been enhanced. Li et al. [15] further expanded the application of FCN in crack detection and proposed a variant of FCN for multi-damage detection of concrete structures, which can simultaneously detect cracks and other structural damages. They optimized the standard FCN structure to improve the detection capability for different types of crack damage. Chu et al. [16] proposed Tiny Crack Net for segmenting small cracks. Tiny-Crack-Net uses an improved ResNet structure as its backbone network to extract deep-level crack features from input images, enhancing its ability to recognize cracks of different scales. Han et al. [17] proposed a CNN-based CrackW-Net to achieve high-precision crack segmentation. CrackW-Net uses a skip-level back-and-forth sampling block structure to solve problems such as continuous crack interruption and misidentification of background discrete noise. To solve the problems of traditional CNN structures in crack detection, such as insufficient ability to capture complex backgrounds and global features, researchers have begun to explore improved methods on top of CNN structures in recent years. Liu et al. [18] added Spatial and Channel Compression and Squeeze-and-Excitation Block between the Batch Normalization layer and Deconvolution layer in U-Net to improve the performance of upsampling blocks for crack feature extraction. Usually, the classification of small image blocks cannot achieve pixel-level accuracy in finding cracks. Jo et al. [19] proposed a composite network combining YOLOv7 and DeepLabv2 for crack detection. YOLOv7 in the composite network is based on a CNN structure. It incorporates a path aggregation network (PAN) and spatial pyramid pooling (SPP) to achieve efficient detection and feature extraction of cracks at different scales. Based on the crack target box detected by YOLOv7, DeepLabv2 enhances the receptive field through Dilated Convolution and utilizes the CNN structure for feature extraction to achieve pixel-level crack segmentation.

With the development of attention mechanisms, segmentation networks based on the Transformer framework [20] have gradually been proposed, especially Vision Transformer (ViT) [21] and its segmentation domain variants, such as Segmentation Transformer (SETR) [22], SegViT [23], which have achieved significant results in the segmentation domain. However, the computational cost of using a pure Transformer architecture in the field of crack segmentation is high. Therefore, researchers do not use a completely pure Transformer architecture for segmentation, and often choose a hybrid structure that combines convolution and the Transformer for segmentation. The hybrid segmentation architecture uses a Transformer structure to compensate for the shortcomings of CNN in global information modeling ability, and improve the generalization ability and robustness of crack detection. Shamsabadi et al. [24] proposed a ViT-based framework for detecting surface cracks in asphalt and concrete. This method uses pure ViT as the backbone feature extraction and employs a CNN-based decoder for end-to-end semantic segmentation. Similarly, Qi et al. [25] introduced a convex variational network based on visual transformers for bridge and pavement defect segmentation. They used the fusion of the ViT structure and dual decoder architecture to refine boundary details and segment crack areas. The first decoder uses a Transformer for semantic feature recovery, and the second decoder uses a Level Set variational method for boundary optimization to ensure the integrity of the crack area. Some studies have explored hybrid CNN-Transformer segmentation networks, combining the advantages of both paradigms. Zhang and Zhang [26] evaluated various hybrid transformer CNN models, including TransUNet [27] and SwinUNet [28], for road crack segmentation. Their results indicate that the architecture based on Swin-Transformer [29] exhibits excellent feature extraction capabilities while maintaining computational efficiency. In another study, Su et al. [30] conducted a comparative analysis of real-time crack detection using CNN, ViT, and hybrid models. They found that SegFormer [31] with MiT-B1 as the backbone achieved the highest segmentation accuracy, while the model based on TopFormer [32] provided the best real-time performance. Their work provides insights into selecting suitable deep learning models for practical crack detection applications. Qu et al. [73] introduced a hierarchical feature fusion and connected-attention architecture, which substantially improved both the accuracy and robustness of crack segmentation. Qu et al. [34] developed a crack detection method based on an attention mechanism and multi-feature fusion, effectively enhancing the representational capacity of crack regions under complex background conditions. Zhou et al. [74] proposed the Potentially Effective Network (PPNet), which integrates enhanced convolution with a dynamic feature fusion strategy, thereby significantly improving the accuracy and generalization performance of crack detection.

Recently, with the emergence of Mamba networks [33] and their visual variants such as VMamba [55], Mamba networks have been introduced in crack segmentation to overcome the

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local receptive field limitations of CNNs and the computational overhead of Transformers. Zuo et al. [35] proposed CrackMamba. They designed a crack segmentation network based on the Mamba structure to address the issues of limited local receptive fields in CNNs and excessive computational overhead in Transformers. This method introduces the Snake Scan module to adapt to the tortuous shape of cracks, and combines the Snake Conv VSS (SCVSS) block to fuse local and global features, improving the accuracy of crack segmentation. Chen et al. [36] proposed VMamba-Net, which is used for crack detection in concrete, asphalt, and masonry. This method uses VMamba as the backbone network, replacing CNN and Transformer, and utilizes linear computational complexity to improve computational efficiency. Meanwhile, the 2D-Select-Scan (SS2D) module is used for four-way feature modeling to enhance crack detection capability.

C. Multimodal Segmentation Method Combining Text

In order to solve the problem that image features cannot fully represent the features of the segmentation target in a single image segmentation task, in medical image segmentation, Fully Convolutional Network and Dilated Fully Convolutional Encoder Decoder Network have been developed for pixel-level segmentation. Essentially, crack segmentation is similar to retinal vessel segmentation. For example, UNet [37], UNet++ [38], and FusionNet [39], which are commonly used for medical image segmentation, can be used for pixel-level crack segmentation. In addition to the information that images can provide about crack features, textual descriptions can also represent additional information in the image. Lüddecke et al. [40] proposed a system for generating image segmentation based on arbitrary text prompts. The addition of text provides additional understandable information for image segmentation. Therefore, medical segmentation has also proposed a dual-modal segmentation method that integrates medical images and text guidance. Dhakal et al. [41] introduced fine-tuning the segmentation model of pretrained medical images and medical language description datasets using Transformer encoders. The VLSM Adapter can guide medical images based on text prompts. In crack segmentation, the crack image itself may not provide sufficient contextual information to understand the characteristics of the crack. By combining descriptive text, the model can better understand and identify different types of cracks and their related features.

III. Methods

In this section, we first introduce the data acquisition process in Section II-A. Then, we present the multimodal segmentation network, TMENet, for crack detection in multiple scenarios in Section II-B. This network incorporates several key modules, including the TCE Module in Section II-C, the Ms-Mamba Module in Section II-D, and the LED module in Section II-E.

A. Data Acquisition

(1) Web crawler. This article uses a web crawler program to download free and publicly available image datasets automatically, and manually filters out images that may involve privacy concerns. (2) On-site filming. 851 crack images were captured using the DJI MAVIC 3 PRO camera provided by the National University of Defense Technology. Partial crack images are shown in Fig. 1. The crack images in the figure all show a certain degree of segmentation difficulty. Fig. 1 (a) is disturbed by oil stains and noise. The crack texture in Fig. 1 (c) is irregular and has many intersections. The color contrast between the crack and the background in Fig. 1 (e) is extremely low, and there is interference from the marking lines in the crack background. The crack in Fig. 1 (g) contains water stains and uneven light and shadow.

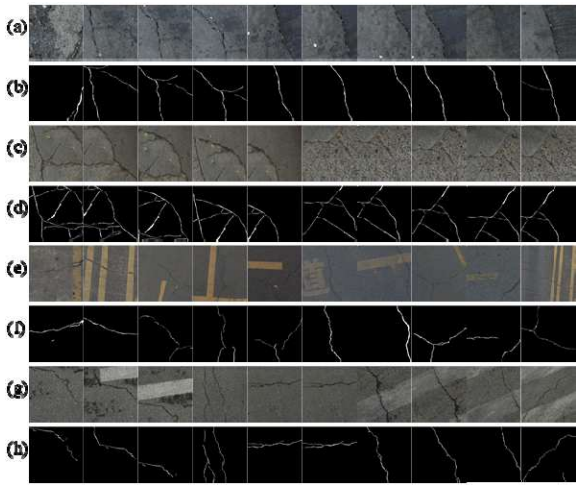

Fig. 1. Partial display of the CampCrack800 dataset. (a), (c), (e), (g): Original images of road cracks with different features; (b), (d), (f), (h): Ground truth of the crack images.

B. Overview of the Multimodal Road Crack Segmentation Network for Complex Scenarios(TMENet)

Clipseg [40] utilizes the powerful language visual joint understanding capability of CLIP and efficiently performs image segmentation through the encoder-decoder structure constructed by Transformer, capable of handling complex multimodal inputs. When using the traditional end-to-end fine-tuning method, Clipseg on a small amount of data, the model will over learn specific features in the training dataset, including noise and background. At this point, the model may not accurately identify cracks when performing crack detection on new or unseen road images. Therefore, we added trainable modules to the frozen Clipseg architecture to fine-tune the modules we added. The fine-tuning module can solve a series of problems caused by Clipseg in road crack segmentation.

As shown in Fig. 2(A), the crack image and crack text will first pass through three channels. The first channel is Text Transformer, which incorporates the TCE Module to encode the input text information. After the text encoding is completed, projection1 is generated and input into the Decoder of Clipseg. The second channel, Vision Transformer, incorporates the Ms-Mamba module to encode the input crack image information. The image encoding information is encoded in the last layer to generate projection2, which is also input into the Decoder of Clipseg. The third channel is the

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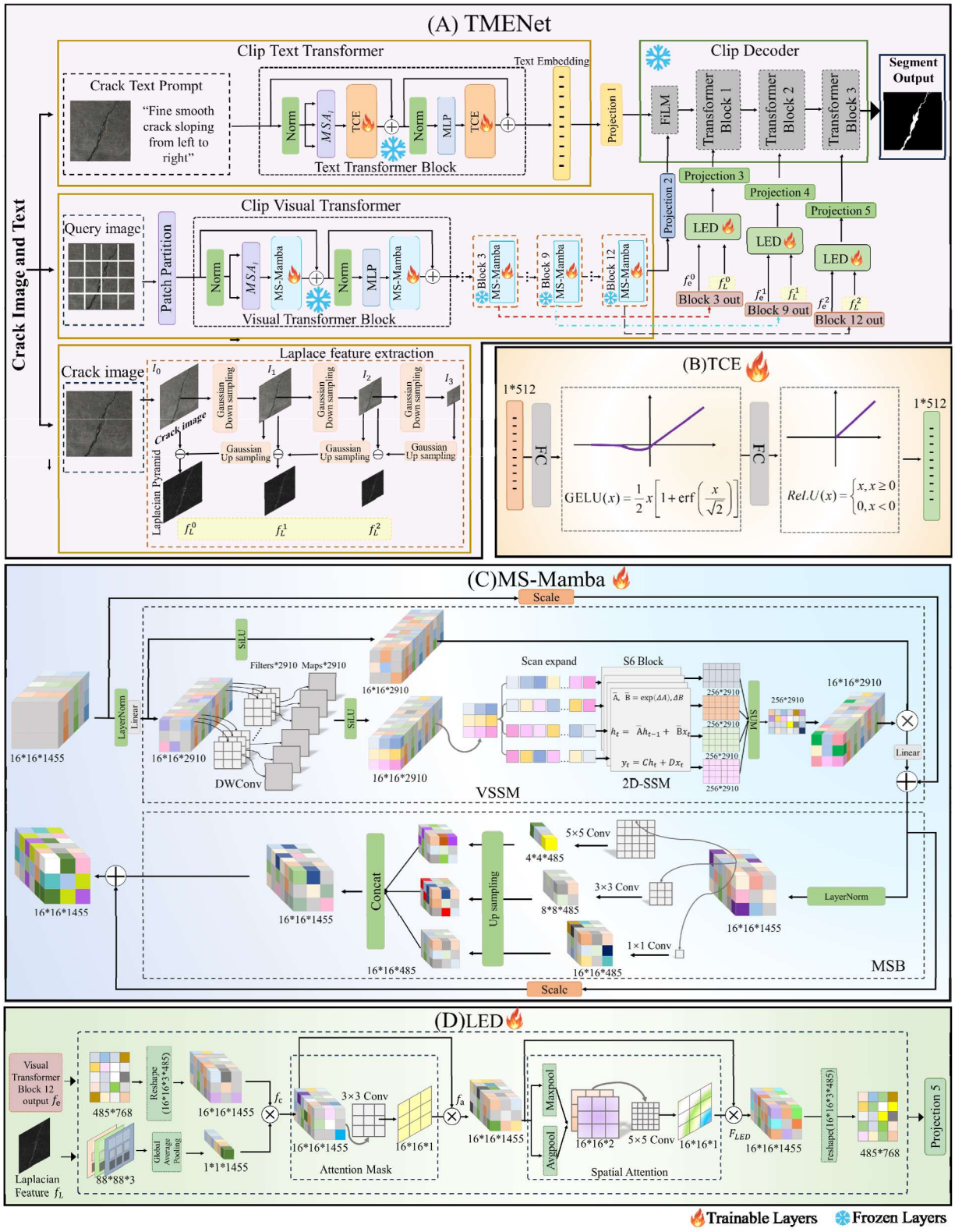

Fig. 2. TMENet framework structure: (A) Overall schematic diagram of TMENet's network structure (B) Structure diagram of TCE module (C) Structure diagram of Ms Mamba module (D) Structure diagram of LED module.

Laplace feature extraction pathway, whose primary function is to extract the Laplacian feature pyramid of the crack image. The LED module integrates Laplacian features with crack image encoding information and locates edge information. Feed the encoded information of the output edge information into the Decoder of Clipseg. Finally, the decoder decodes the information from three different paths to generate road segmentation results.

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C. Textual Crack Enhancer Module(TCE Module)

The description samples of road crack data trained by Clipseg before fine-tuning are limited, and the model may not be able to learn how to handle uncommon crack quantification or specific crack description methods. Therefore, the Clipseg model cannot accurately parse the descriptive information of cracks in text and map this information to the corresponding areas of crack images. Consequently, we introduced the TCE module to fine-tune the Text Transformer's ability to adapt to text tasks in road cracks.

As shown in Fig. 2(B), we introduced TCE Modules into the Text Transformer Block of the Clipseg model. Each self-attention module contains two residual blocks. Therefore, each self-attention layer uses two TCE Modules. In (B), MSA_l a multi-head self-attention block is represented, and MLP stands for Multilayer Perceptron. The TCE Module is a nonlinear projection block that adapts the representation of the pre-trained model to downstream tasks without changing its parameters, thereby making the representation learned by the pre-trained model applicable to other tasks. As shown in equation (1), this is the formula for the TCE Module:

$$f' = Adapter(f) = f + \sigma(\psi(f \cdot W_1) \cdot W_2) \tag{1}$$

Among them, f is the representation learned by the pre-trained model, f' is the adaptive feature, W_1 and W_2 are learnable weights. ψ and σ are nonlinear activation functions, where ψ represents GeLU and σ represents ReLU. The TCE Module is designed to fine-tune the encoder by inserting the TCE Module before each attention layer while keeping the pre-trained model parameters frozen. The TCE Module is deployed in the self-attention layer of the image encoder to finely adjust the internal representation of the model with smaller hidden dimensions. The addition of the TCE Module enables the Clipseg model to quickly adapt to road crack segmentation tasks while maintaining its rich knowledge and generalization ability obtained from pre-training in new tasks.

The TCE module exhibits structural similarities to the conventional SE module, as both employ a lightweight nonlinear mapping composed of a linear layer–activation–linear layer sequence. Nevertheless, their functional roles differ substantially. The SE module utilizes global average pooling to extract channel statistics and applies a two-layer nonlinear mapping to achieve adaptive channel weighting. Essentially, it functions as a channel attention mechanism aimed at refining channel representations of image features. In contrast, the TCE module is not concerned with channel weighting. Instead, it is positioned before the Transformer's self-attention layer, where it employs a lightweight nonlinear projection to adaptively adjust pretrained text feature vectors in the feature space. Consequently, while the SE module is better suited for addressing channel weight distribution in unimodal image tasks, the TCE module is specifically designed for cross-modal tasks, with an emphasis on enhancing semantic understanding and feature alignment.

D. Multi-Scale Mamba Module(Ms-Mamba Module)

Clipseg encodes images based on Vision Transformer [21]. In the ViT encoder, the self-attention mechanism is used to capture global dependency relationships. However, in tasks that require fine feature recognition, the limitations of ViT in processing local pixel information cannot be ignored. However, the self-attention mechanism of the Transformer can theoretically cover all input parts. However, in practical applications, attention distribution is often uneven, leading to the neglect of some important local information. This uneven distribution may be concentrated in a particular area of the image while ignoring other parts. To address this issue, we have introduced a multiscale Mamba module based on the VSSM module [54] in the image encoder section.

The structure of the Ms Mamba module is shown in Fig. 2 (C). Similar to the TCE Module in the previous section, they are both added after the multi-head attention and the MLP of the Transformer Block. Next, we will provide a detailed introduction to the structure of the Ms Mamba module.

To convert the two-dimensional features into the three-dimensional format required by the Ms Mamba module, we must ensure the total number of features remains unchanged. The original image size processed by the Transformer Block is 352 × 352 pixels, divided into 16 × 16 pixel blocks. Each block is flattened across three RGB channels to form a 768-length vector, resulting in 22 × 22 blocks. After adding positional encoding, the image blocks are organized into 485 independent feature dimensions, aligning with the output size of the Transformer Block. In the feature rearrangement process, each 16 × 16 pixel block serves as a processing unit, integrating information from other dimensions into channel data. The 485 × 768 feature map is then rearranged into 16 × 16 × 1455 high-dimensional features. This transformation not only meets the input format requirements of the Ms Mamba module but also preserves spatial information, enriching the contextual information for subsequent feature learning.

Next, we will provide a detailed introduction to the structure of the Ms-Mamba Module. Firstly, the Ms-Mamba Module generates two feature maps with the exact size of 16 × 16 × 2910 through LayerNorm and linear transformation, and inputs them into two branches for feature extraction. The main branch captures long-range spatial dependencies through VSSM blocks. VSSM Block applies depthwise separable convolution and SiLU activation functions to improve computational efficiency and nonlinear expression ability, thereby enhancing feature expression ability while ensuring efficient computation. By combining the 2D-SSM layer [55], the two-dimensional spatial information is better utilized. 2D-SSM converts two-dimensional image data into one-dimensional sequences by scanning images along four directions. Then it merges these sequences back into a two-dimensional structure through summation and reconstruction operations. The introduction of VSSM significantly improves the model's sensitivity to multiscale information and its ability to understand global context. The VSSM module adopts a dual-branch structure to enhance processing efficiency. The calculation formula for the main branch is:

$$X_1 = \text{LN}(\text{2D-SSM}(\text{SiLU}(\text{DWConv}(\text{Linear}(X))) \tag{2}$$

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The other activation branch is used for feature extension, and the process is as follows:

$$X_2 = SiLU\,(Linear(X)) \tag{3}$$

Subsequently, multiply the outputs of these two branches to obtain the output features of the VSSM Block:

$$X_2 = SiLU\,(Linear(X)) \tag{4}$$

In addition, we introduced a learnable scaling factor to control the information in skip connections and form residual branches. The state space feature formula generated by combining the residual branch with the VSSM branch is:

$$\text{X'} = VSSM(LN(X)) + s \cdot X \tag{5}$$

Among them, s is a learnable scaling factor used to adjust the amount of information in residual connections to control the degree of preservation of input feature X in the Ms-Mamba Module.

Due to SSM processing two-dimensional feature maps into one-dimensional token sequences, the number of neighboring pixels is significantly affected by this flattening strategy. Some adjacent pixels in the two-dimensional feature map may be far apart in the one-dimensional token sequence, which can result in local information loss. Maintaining the integrity of local neighborhoods is crucial when processing data with highly structured spatial information. To this end, the Ms-Mamba Module introduces additional local convolutions to restore neighborhood similarity after obtaining state space features. In order to effectively integrate image details of different scales, we adopted multi-scale composite convolution. Specifically, we normalize the features using LayerNorm. Then, the normalized features are processed in parallel using 1×1, 3×3, and 5×5 convolution kernels to extract multi-level spatial information. Unlike conventional multi-scale convolution (e.g., Inception), which applies different kernel sizes (1×1, 3×3, 5×5) at the same resolution for parallel feature extraction, our design extends the concept of multi-scale convolution. Specifically, the 5×5 convolution is not used for channel expansion but serves as a compression and refinement operation, extracting more representative features under a larger receptive field. In contrast, the 1×1 and 3×3 convolutions preserve and enrich local details from fine edges to neighborhood structures. The combination of 1×1, 3×3, and 5×5 convolutions can therefore be regarded as a hierarchical feature screening process, where features of different levels of importance are selected and fused. This extended form of multi-scale convolution is particularly beneficial for reorganizing the token features output by VSSM, as it compresses redundant information while reinforcing local detail and cross-scale context, ultimately achieving a more balanced representation between global and local features.

In addition, similar to the VSSM module, the Ms-Mamba Module also introduces residual branches controlled by scaling factors. The complete formula for this process is:

$$X_{out} = Conv_{MC}(LN(X') + s' \cdot X' \tag{6}$$

Among them, it represents a composite convolution operation that integrates multi-scale composite convolution, internally processing 1×1, 3×3, and 5×5 convolution kernels in parallel and merging the results. $s' \cdot X'$ is the residual connection part, and s' is the learnable scaling factor.

The Ms-Mamba Module significantly enhances the model's ability to recognize multiscale and local details in images by integrating multi-scale composite convolution and residual connections. This design effectively addresses the common problem of uneven attention distribution in ViT-based self-attention mechanisms, thereby avoiding the loss of key local information. The Ms-Mamba Module, which integrates VSSM and 2D-SSM, not only enhances the model's ability to capture long-range spatial dependencies but also deeply extracts and restores potentially forgotten local information. Residual connections maintain the integrity of information, enabling the Ms-Mamba Module to enhance attention to important details while maintaining the richness and accuracy of input features.

E. Laplacian Edge Detection Module(LED Module)

In complex road images containing cracks, there may be many elements similar to cracks, such as road markings, surface textures, etc. Similar characterization features make it challenging to capture cracks using simple threshold segmentation. The edges of cracks usually contain high-frequency components in the image. The Laplacian pyramid can capture high-frequency information of second-order rapid changes in an image. Therefore, in crack detection, the Laplacian operator can effectively help identify cracks or fracture edges on the surface of the object. This article designs an edge detection LED module that integrates Laplace features. The LED module integrates Transformer depth features and Laplacian edge features, and then dynamically adjusts the fused features to obtain the fused depth edge features.

As shown in Fig. 2(D), the LED module receives two main inputs. The first input is the traditional visual feature f_{e} extracted from the original road crack image. Before being transmitted to the LED module, f_{e} extracts and integrates information from the input raw RGB road crack images through the self-attention mechanism in the Vision Transformer Block. The second input is the high-frequency edge feature f_L extracted from the Laplacian feature pyramid [56]. The generation process of the Laplacian feature pyramid presented in the Laplacian feature extraction channel in Fig. 2 (A) is shown.

Firstly, the original crack image data I_0 is treated as the bottom layer of a Gaussian pyramid. For each layer I_n, a Gaussian filter is used to smooth the image of that layer, followed by downsampling. During downsampling, each dimension is halved to generate the next layer $I_{\{n+1\}}$. Each layer L_{n} of the Laplacian pyramid is calculated by subtracting the image upsampled and Gaussian smoothed by the next layer I_{n} from the current Gaussian $I_{\{n+1\}}$. The specific formula can be expressed as:

$$L_n = I_n - d\left(g(I_{\{n+1\}})\right) \tag{7}$$

d is the Gaussian filter convolution operator, and g represents a $2\times$ upsampling operation. In this way, each layer L_{n}

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of the Laplacian feature pyramid represents the high-frequency components of the crack image at different scales.

The LED module performs Global Average Pooling on f_L after receiving high-frequency edge features f_L from the feature layer in the Laplacian pyramid. Global Average Pooling transforms the input 3-channel Laplacian pyramid image into a $1 \times 1 \times 1455$ feature layer. The Transformer Block input to the LED module's 485×768 feature layer transforms the scale to $16 \times 16 \times 1455$. This transformation is similar to the transformation process in the Ms Mamba module. Essentially, this is reshaping the image size into a feature with a length and width of 16 and a channel of 1455 while ensuring that the features remain unchanged. And the size of each channel is 16×16, which corresponds to the architecture's initial use of Patch Partition to divide the image into 16×16 image blocks. Then multiply the restored Laplacian features with the features of the Transformer Block to obtain a fused feature module of $16 \times 16 \times 1455$.

$$f_c = f_L \times f_e \tag{8}$$

where f_c represents the fused output feature. Subsequently, in order to better capture local edge features and related spatial information, we use a 3×3 convolution kernel to process the fused feature f_c. The convolutional layer is configured with padding and a step set to 1 to ensure that the attention mask A has the same size as the original image. The generated attention mask A is used to modulate the edge features in the original feature map. Then, combining mask A with the fused feature map enhances the depth edge information and suppresses unimportant background noise and details. The formula for generating the attention mask result is as follows:

$$\begin{aligned} f_a &= A \times f_c \\ &= \sigma(Conv_{3\times3}(f_c)) \times f_c \end{aligned} \tag{9}$$

Finally, the LED module integrates a spatial attention mechanism (SA) to enhance the expression ability of edge features in complex scenes. SA learns the importance of each pixel in the fused features. Firstly, f_a obtains the maximum and average feature maps through Maxpool and Avgpool. Then, the maximum feature map and the average feature map are added and convolved by 5×5 to obtain the spatial attention weight map. We multiplied the spatial attention weight with f_a to obtain the edge feature F_{LED} that has undergone spatial attention. The formula for generating F_{LED} is as follows:

$$\begin{aligned} F_{LED} &= SA(f_a) \times f_a \\ &= \sigma(Conv_{5\times5}((M\text{ax}(f_a) + Avg(f_a)) \times f_a \end{aligned} \tag{10}$$

Among them, $Conv$ represents generating a new attention weight map through a two-dimensional convolution operation. Max obtained the maximum feature map of f_a. Avg obtained the average feature map of f_a. F_{LED} represents the final output feature map of the LED module. However, it is worth noting that the size of F_{LED} is $16 \times 16 \times 1455$, and it needs to be expanded to 485×768 to participate in the next module's operation.

The feature F_{EGA} processed by the spatial attention mechanism is a refined feature of deep edge features that have been reweighted and corrected in the spatial dimension. These features can better capture edges and details in images while suppressing interference from complex backgrounds, thereby improving the performance of the model in handling complex scenes. Therefore, LED modules can significantly distinguish cracks from background noise, improving the accuracy of crack segmentation.

IV. Experiment

A. Experimental Environment and Settings

In our experiment, the proposed TMENet was trained using the AdamW optimizer [57] with an initial learning rate of 0.001, momentum of 0.9, and weight decay of 0.001. The learning rate scheduler is set to ReduceLROnPlateau, with a factor of 0.3 to reduce the learning rate and a patience parameter of 5. The maximum training period for all models is 300 epochs. But if there is no progress in DSC (%) verification for 20 consecutive epochs, the training will be stopped. All loss functions used in networks are a mixture of Dice and cross-entropy loss, with Dice and cross-entropy loss weights of 1.5 and 1.0, respectively. In order to ensure the accuracy and fairness of the experiment, two evaluation indicators were used, including the mIoU coefficient and the Dice similarity coefficient. All models are based on the PyTorch platform and trained on an NVIDIA RTX 4090 GPU.

B. Best Text Length Verification Experiment

In our self-constructed dataset, we assigned textual labels of varying lengths (P0–P9) to each crack image. These labels were progressively refined, with P5 emerging as the optimal balance point. Unlike shorter or overly simplistic descriptions, P5 not only captures the geometric and textural attributes of crack foregrounds—such as fine cracks, low contrast, and rough edges—but also incorporates background interference factors, including water stains, markings, and noise. This combination enables the most effective correspondence between the visual and textual modalities.

For public datasets, we adopted P5 as a general textual prompt, formulating descriptions based on typical crack and background characteristics without the need for image-by-image annotation. When encountering images with substantially different global features, we adjusted the descriptions accordingly and further enriched them with details such as crack distribution. Examples of the final textual labels are provided in the first column of Table I, where P0–P5 illustrate the stepwise refinement process and P7–P9 serve as supplementary, more detailed descriptions. The segmentation outcomes across different text lengths are also reported in Table I.

The segmentation results of TMENet confirm that P5 yields the best performance. Interestingly, longer textual inputs do not necessarily improve segmentation accuracy. While P5 delivers precise and contextually relevant guidance—including crack geometry, edge roughness, texture, contrast, and environmental interference—descriptions in P6–P9 introduce abstract or vague terms that provide less reliable cues, ultimately degrading segmentation quality. Consequently, in subsequent experiments, we consistently employed P5-like prompts for

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TABLE I
COMPARISON OF SEGMENTATION RESULTS FOR TEXT LABELS OF DIFFERENT LENGTHS IN CAMPCRACK800

Text labels of different lengths	IOU	Dice
P0:" "	0.8923	0.9421
P1: "Cracks"	0.8781	0.9341
P2: "Multiple jagged-edged black cracks"	0.8814	0.9359
P3: "Multiple jagged-edged black cracks extending vertically"	0.8910	0.9415
P4: "Multiple vertical black cracks with jagged uneven edges"	0.8862	0.9389
P5: "Centered multiple vertical black cracks with jagged uneven edges and branches"	**0.9066**	**0.9507**
P6: "The multiple vertical black jagged edge cracks in the center of the image are lines of different widths without branching."	0.8745	0.9318
P7: 0: "Road cracks are lines of irregular shape on the surface." 1: "Road cracks are separations on the surface of asphalt or concrete that can be linear." 2: "The width and depth of road cracks usually vary." 3: "Road cracks are cracks from the surface to deeper levels, sometimes filled with debris." 4: "Road cracks can be lines of varying width without branches."	0.9053	0.9495
P8: 0: "Centered multiple vertical black cracks with jagged edges are lines of irregular shape on the surface." 1: "Centered multiple vertical black jagged-edged cracks are separations on the surface of asphalt or concrete that can be linear." 2: "Centered multiple vertical black jagged-edged cracks usually vary in width and depth." 3: "Centered multiple vertical black jagged-edged cracks are from the surface to deeper levels without debris." 4: "Centered multiple vertical black jagged-edged cracks can be lines of varying width without branches."	0.8947	0.9435
P9: 0: "Centered multiple vertical black cracks with jagged uneven edges and branches are lines of irregular shape in the image center" 1: "Centered multiple vertical black jagged-edged cracks in the image center are separations on the concrete surface that can be linear." 2: "The centered multiple vertical black jagged-edged cracks in the image center vary in width and depth." 3: "The centered multiple vertical black jagged-edged cracks in the image center are from the surface to deeper levels without debris." 4: "The centered multiple vertical black jagged-edged cracks in the image center can be lines of varying width without branches."	0.8981	0.9456

public datasets. Such prompts sufficiently convey the global characteristics of crack images and, as demonstrated, provide effective guidance for distinguishing foreground from background features, thereby enhancing crack feature extraction.

It is worth emphasizing that, after establishing the P5 descriptive paradigm in our self-constructed dataset, we did not perform additional per-image textual annotations in later experiments with public datasets, as this would severely impact efficiency. Instead, we designed three to four variations of textual prompts—each expressing the same crack-related foreground and background features in slightly different ways—based on the P5 paradigm. Each image was paired with one of these prompts during training. This approach avoided the inefficiency of manual annotation while still delivering robust multimodal guidance to improve crack segmentation. Furthermore, when datasets included images with markedly different global characteristics, we refined the prompts accordingly and supplemented them with additional descriptions of crack distribution to maintain the effectiveness of textual guidance.

C. Text Information Validity Verification Experiment

To further assess the contribution of textual information introduced by the TCE module in crack segmentation, we curated several representative complex crack scenarios from our self-constructed CampCrack800 dataset: (1) cracks with irregular or rough edges, (2) fine cracks with low contrast, (3) cracks affected by background interference such as stains or road markings, and (4) cracks characterized by complex, multi-intersecting textures. For each of these cases, we employed P5-type textual descriptions, which incorporate both foreground crack attributes—such as morphology, edge roughness, texture complexity, and contrast—and background disturbance factors, including stains, shadows, and marking lines.

In our experiments, we compared segmentation performance between two models: Clipseg without TCE and Clipseg enhanced with TCE (P5). To gain deeper insights into how textual information influences the model's internal feature learning, we further conducted visualization analysis using Class Activation Maps (CAMs). Unlike evaluations that rely solely on segmentation outputs, CAMs reveal the regions that the model attends to during prediction. This allows us to determine whether the model truly focuses on the cracks themselves and whether the textual prompts effectively guide the model to suppress irrelevant background interference.

The visualization results further highlight the contribution of the TCE module to the effectiveness of textual guidance. For cracks with rough edge morphology (Fig. 1-1 and 1-2), the Clipseg model without TCE generated dispersed activation regions, with responses often extending to rough textures beyond the actual crack boundaries. In contrast, Clipseg enhanced with TCE exhibited more precise activations along the main body and edges of the cracks, indicating that textual prompts successfully guided the model to capture critical edge features.

For fine cracks under low-contrast conditions (Fig. 2-1 and 2-2), the baseline model was easily affected by background gray-level variations, resulting in weak responses. With TCE integration, however, CAM visualizations revealed

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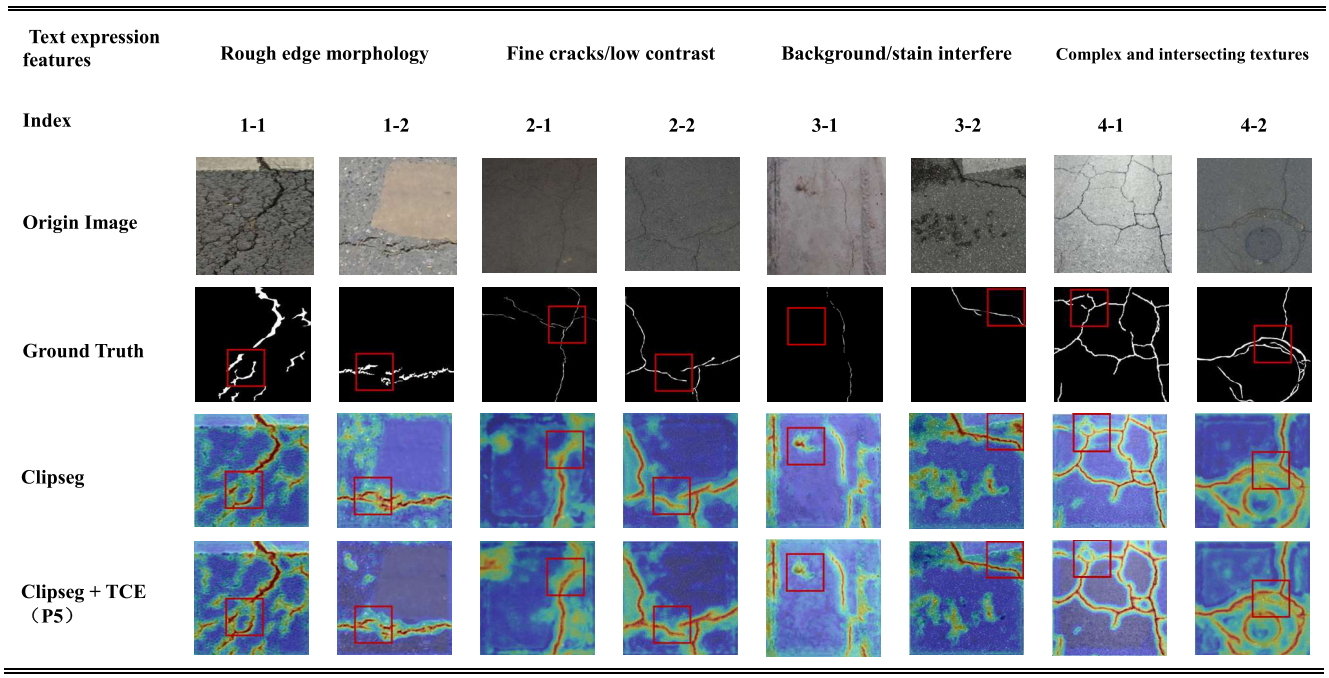

Fig. 3. Verification results of text information validity based on CAM.

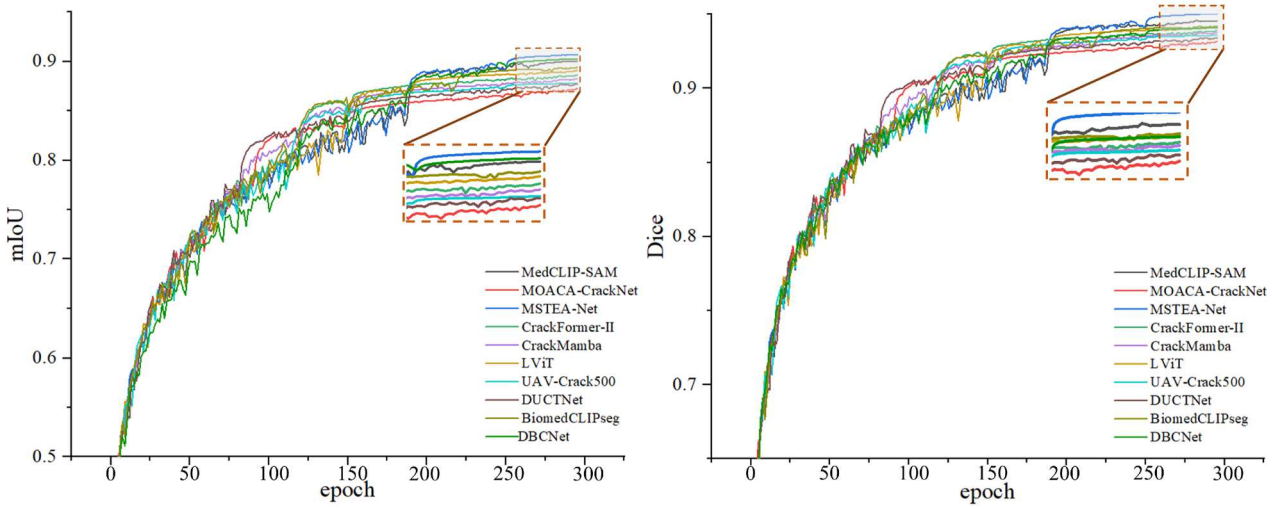

Fig. 4. Network comparison test graph in CampCrack800.

significantly stronger activations along the crack regions, showing that the model was better able to concentrate on subtle crack structures that are otherwise difficult to detect. In cases with background interference, such as stains (Fig. 3-1 and 3-2), the Clipseg model produced erroneous activations in non-crack regions. The introduction of TCE effectively suppressed these false responses, allowing the model to focus more accurately on the true crack areas. Finally, in complex scenarios characterized by multiple intersecting textures (Fig. 4-1 and 4-2), Clipseg exhibited substantial confusion between cracks and background patterns. With TCE, however, the crack activations became more continuous and coherent, while the influence of background textures was significantly reduced.

Collectively, these results demonstrate that the TCE module strengthens the alignment between textual and visual representations, enabling the model to achieve a more accurate semantic understanding of cracks and thereby enhancing multimodal fusion. This not only leads to measurable improvements in segmentation accuracy but also provides visual evidence of the model's more rational and focused attention.

D. Ablation Experiment

To ensure the fairness of the experimental results and the stability of the network, this section conducted ablation experiments on three methods using a self-made dataset and other public road cracks, AsphactCracks300 [58], Crack9K [59], and CRKWH100 [60]. In this experiment, TCE,

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TABLE II
RESULTS OF ABLATION EXPERIMENT

Dataset	CampCrack800		AsphaltCrack300	
Network	**mIoU**	**Dice**	**mIoU**	**Dice**
Clipseg	0.7918	0.8684	0.8059	0.9096
TCE	0.8168	0.8921	0.8493	0.9144
MsMamba	0.8620	0.9104	0.8735	0.9152
LED	0.8526	0.9137	0.8604	0.9220
TCE&MsMamba	0.8978	0.9476	0.9024	0.9377
TCE& LED	0.8803	0.9435	0.9094	0.9353
MsMamba & LED	0.9014	09502	0.9179	09407
TMENet	**0.9066**	**0.9507**	**0.9103**	**0.9447**
Dataset	**Crack9K**		**CRKWH100**	
Network	**mIoU**	**Dice**	**mIoU**	**Dice**
Clipseg	0.7629	0.7538	0.6985	0.7997
TCE	0.7729	0.7952	0.7037	0.8125
MsMamba	0.7773	0.7913	0.7509	0.8175
LED	0.7890	0.8009	0.7425	0.8180
TCE&MsMamba	0.8021	0.8254	0.7885	0.8414
TCE& LED	0.8174	0.8495	0.7724	0.8464
MsMamba & LED	0.8235	0.8363	0.7997	08574
TMENet	**0.8273**	**0.8499**	**0.7926**	**0.8531**

TABLE III
EXPERIMENTAL DATA METRICS FOR TMENET GENERALIZATION COMPARISON

Dataset	Rank	CrackForest		Rank	BC-DUNet	
Network		**IOU**	**Dice**		**IOU**	**Dice**
TMENet	1	**0.7429**	**0.8479**	1	**0.8671**	**0.9346**
MedCLIP-SAM	3	0.7253	0.8242	2	0.8512	0.9285
BiomedClipseg	4	0.7137	0.8279	7	0.8424	0.9162
DBCNet	2	0.7414	0.8092	3	0.8499	0.8954
LVIT	5	0.6874	0.7794	4	0.8480	0.9168
CrackMamba	6	0.6801	0.7903	5	0.8459	0.9218
DUCTNet	7	0.6944	0.7946	8	0.8362	0.9070
CrackFormer-II	8	0.6619	0.7385	6	0.8454	0.9107
MOACA-CrackNet	9	0.6553	0.7526	10	0.8021	0.8159
UAV-Crack500	10	0.6417	0.7354	9	0.8284	0.8551
Dataset	**Rank**	**CrackSegNet**		**Rank**	**SCCDNet**	
Network		**IOU**	**Dice**		**IOU**	**Dice**
TMENet	1	**0.7536**	**0.8461**	1	**0.8236**	**0.9180**
MedCLIP-SAM	2	0.7436	0.8384	5	0.8052	0.8964
BiomedClipseg	5	0.7260	0.8266	4	0.8084	0.8953
DBCNet	3	0.7412	0.8252	3	0.8096	0.8724
LVIT	4	0.7264	0.8253	8	0.7870	0.8773
CrackMamba	6	0.7162	0.7923	2	0.8096	0.9028
DUCTNet	8	0.6808	0.7744	6	0.7944	0.8886
CrackFormer-II	7	0.7080	0.8131	7	0.7883	0.8743
MOACA-CrackNet	9	0.6720	0.7587	9	0.7664	0.8552
UAV-Crack500	10	0.6402	0.7193	10	0.7483	0.8334

Ms-Mamba, and LED were added to TMENet in a sequential combination. Table II shows the experimental results.

The experimental results in Table II indicate that the three methods have different directions of improving the index coefficients on four different datasets. Among them, TMENet (TCE&Ms-Mamba&LED) achieved the highest performance in all evaluation metrics, with an improvement of approximately 6%-9% compared to the original network on all four datasets. From these experiments, it can be demonstrated that the three methods proposed in TMENet are effective.

E. Comparison Experiments With Crack Segmentation Methods

This experiment selected eight representative crack segmentation networks, covering different modal types and network structures. This includes three types of crack segmentation networks based on CNN skeleton and single image modality as input: UAV-Crack500 [62], DUCTNet [45], and MOACA-CrackNet [61]; A road crack segmentation network based on Transformer architecture, CrackFormer-II [44], and road crack segmentation networks called CrackMamba [35] and DBCNet [68] based on Mamba skeleton. In addition, we also selected three segmentation networks based on the Transformer architecture that fuse text and image bimodal information: LViT [63], BiomedCLIPSeg [64], and MedCLIP-SAM [40]. Fig. 4 shows the testing curves of ten networks on the CampCrack800 dataset. According to the results of the test curve, TMENet showed better testing performance on the CampCrack800 dataset. Although its convergence time is relatively late, the convergence accuracy is high, which

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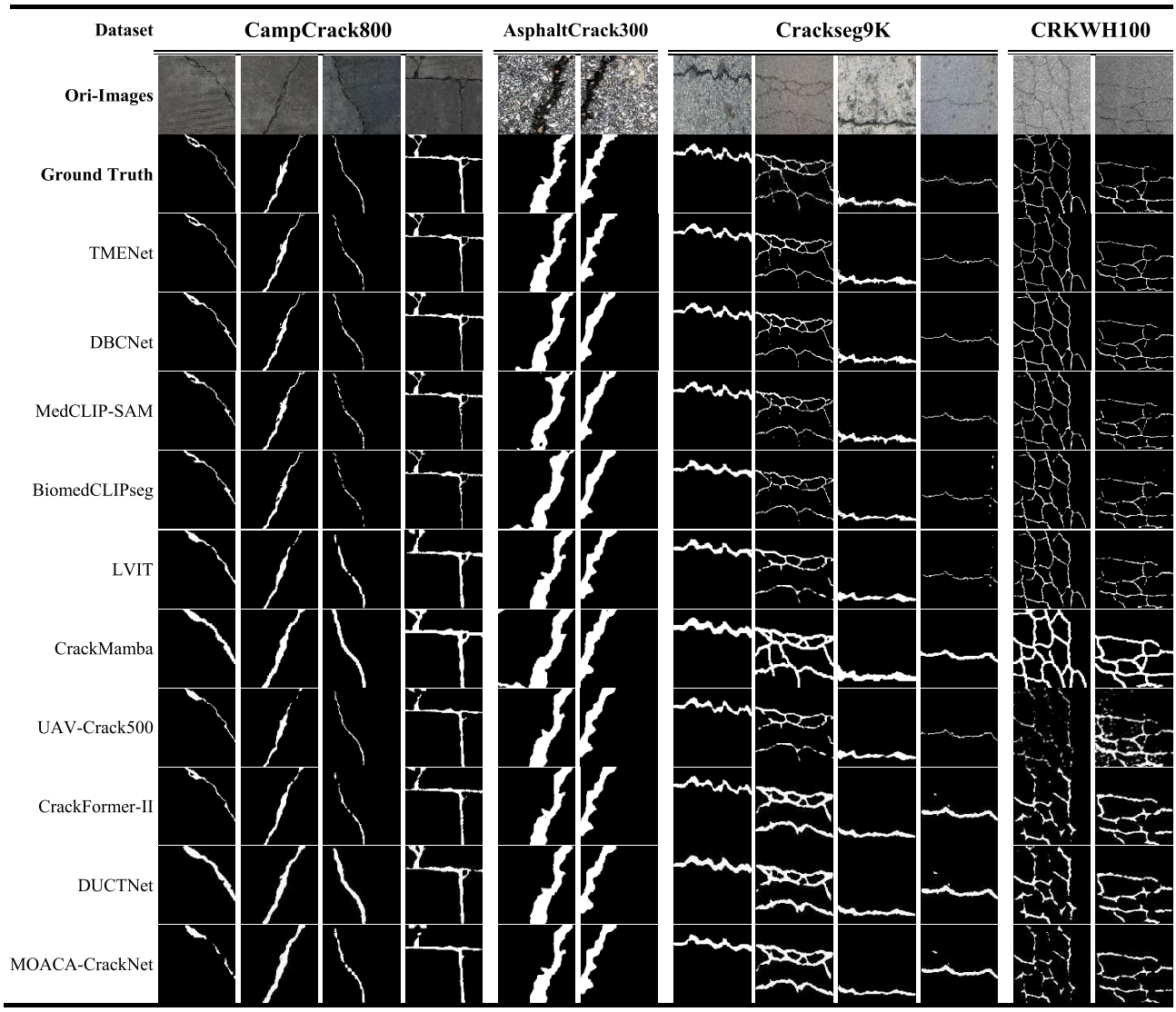

Fig. 5. Display of segmentation results from ten networks on the CampCrack800, AsphaltCrack300, Crackseg9k, and CRKWH100 datasets.

indicates that our TMENet model has a large capacity and strong expression ability, and requires more time to gradually learn the feature representations of complex cracks and crack-like objects.

In addition, we conducted nine network segmentation comparison experiments on three public datasets (Asphalt Crack300, Crackeseg9K, CRKWH100). The partial segmentation results of all networks are shown in Fig. 5.

It can be seen that the segmentation results of the TMENet network are closer to the real ground situation. It has better performance in complex backgrounds and can also achieve accurate segmentation in complex backgrounds. Our network's segmentation accuracy for small cracks and complex background cracks is closest to the cracks given in real ground conditions. Other models, such as CrackMamba, CrackFormer-II, DUCTNet, and MOACA-CrackNet, suffer from insufficient refinement of the segmented cracks due to the lack of boundary perception, large receptive fields, and other issues. Our improved network significantly enhances its ability to locate and extract fine cracks through a hybrid improvement strategy, such as edge compensation and multiscale extraction. Secondly, from the segmentation results, it can be seen that our network did not generate misclassifications for the last set of images with water stains in Crackeseg9K. This indicates that he has been able to correctly distinguish between dark background interference features and true crack features based on human growth habits and the common sense of cracks.

F. Generalization Experiments

To test the generalization performance of TMENet, this section selects four authoritative public datasets for experimentation. In order to evaluate the effectiveness of the model on multiple types of samples, this section conducted generalization experiments on cracks of different categories. The experimental dataset for generalization experiments includes the road crack dataset, CrackForest [65]. Crack dataset SCCDNet [66]. The building cracks in SCCDNet exhibit complex and diverse textures—the bridge crack image dataset BC-DUNet [48]. The image size of the bridge crack data in BC-DUNet is relatively large. Large crack images pose

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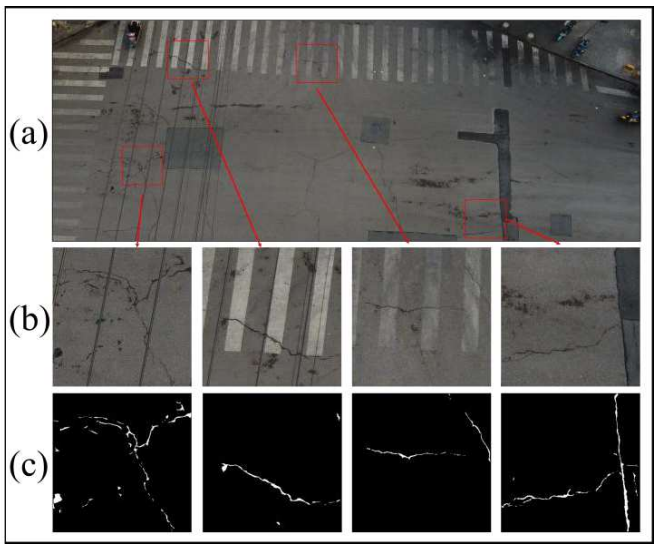

Fig. 6. Real road testing: (a: drone on-site captured images; b: cropped 352 × 352 road segmentation input images; c: TMENet actual segmentation results).

a higher challenge to segmentation accuracy. The tunnel crack dataset, CrackSegNet [67], contains a large amount of interfering substances such as paint and stains in the crack background, which further increases the difficulty of accurate segmentation. Table III shows the segmentation results of the generalization experiments on these four public datasets. Our TMENet achieved the best segmentation performance on road, bridge, building, and tunnel crack datasets, and experiments have shown that the crack segmentation network has strong generalization ability. This indicates that the network truly understands the growth characteristics of the object itself, rather than simply fitting the dataset to road cracks.

G. Actual Road Test

In order to evaluate the practical performance of TMENet, this article used high-resolution devices to collect high-difficulty road crack images. Fig. 6 (a) represents the original collected road image. This includes noise interference such as road route background, black and gray stains, and road object background that interfere with crack segmentation. (b) is the cropped crack image of 352 × 352, and (c) is the segmentation result obtained using a trained network. From the segmentation results, although TMENet has a low false alarm rate, it can accurately segment noticeable cracks in complex scenes, meeting the requirements of road crack segmentation in complex backgrounds.

V. Discussion

TMENet, as an intelligent crack segmentation network that integrates text guidance and multiscale edge enhancement, can significantly promote intelligent transportation systems. Firstly, in terms of road safety detection, TMENet can achieve all-weather and fully automatic road safety monitoring by automatically detecting and segmenting road cracks, significantly improving the efficiency and accuracy of road maintenance. Secondly, in road quality assessment, TMENet can not only accurately identify the types and forms of cracks, but also integrate multimodal information through text-guided modules to enhance a comprehensive understanding of crack characteristics and assist road maintenance departments in conducting scientific quality assessments. Thirdly, based on the crack detection results of TMENet, the intelligent transportation system can automatically generate graded repair strategies, achieving intelligent and precise road maintenance. Fourthly, in the field of intelligent driving and autonomous driving, the crack detection information provided by TMENet can be integrated into the autonomous driving perception system to detect and avoid road cracks in real time, ensuring driving safety. Finally, in urban infrastructure monitoring, TMENet can monitor the structural health of bridges, tunnels, and urban roads, building a comprehensive road health monitoring system for intelligent transportation systems. Through the above applications, TMENet can effectively enhance the intelligence level of road safety monitoring, autonomous driving support, and infrastructure health assessment in intelligent transportation systems, and build a safe, efficient, and sustainable intelligent transportation ecosystem.

Although TMENet has achieved good results in fixed-scale crack image segmentation tasks, the model still has two directions that urgently need further research and improvement in practical engineering applications and prospects.

Firstly, regarding the issue of automatic crack localization in high-resolution scenes. In real-world segmentation experiments, the input of the model mainly relies on manually cropped image segments. Although manually cropped images are convenient for training and evaluation, they cannot cover the entire process of crack detection in real road environments. In images captured by drones or high-resolution camera systems, cracks are often subtle areas in the image, and relying on manual screening will significantly limit the practicality of the model. Therefore, we believe it is necessary to build a fully automated process that includes "coarse-grained crack localization and fine-grained segmentation" in the future. By introducing preliminary region candidate detection algorithms to locate suspected crack areas and then using TMENet for fine-grained super-resolution segmentation, the overall detection efficiency and system automation capabilities are further improved, making it more suitable for large-scale road inspections, intelligent transportation, and urban infrastructure maintenance scenarios.

Secondly, regarding the further expansion of crack semantics in text guidance. The current text description mainly focuses on the visual feature dimensions of cracks in images, such as color, direction, position, and edge morphology. Although it effectively improves the model's attention to key information, in practical applications, the cause and type of cracks have a decisive impact on crack morphology. Different types of cracks (such as fatigue cracks, temperature cracks, settlement cracks, etc.) often exhibit different topological structures, texture distributions, and evolutionary trends in images. However, limited by the establishment of the dataset, this article only conducted preliminary attempts and explorations on text for crack segmentation. We are also gradually trying to introduce crack cause analysis and

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type knowledge into text guidance, enriching the expression dimensions and engineering value of the text by integrating the crack naming system, cause explanation, structural semantics, and other contents in the field of civil engineering. At the same time, introducing domain-specific terminology, causal reasoning information, and related engineering knowledge is expected to further enhance the model's understanding of crack semantic information and achieve an automated crack detection system with professional explanatory power and engineering guidance value.

In recent years, crack segmentation approaches that integrate Mamba with CNNs have emerged as a promising research trend. For example, DCCM-Net [69] leverages the complementary strengths of the two paradigms—utilizing Mamba's capability for long-range dependency modeling alongside CNNs' proficiency in local feature extraction—while further introducing a boundary enhancement mechanism to improve segmentation accuracy. This line of work underscores the significant potential of combining Mamba with conventional architectures for crack detection. In contrast, our proposed TMENet advances this direction by integrating textual guidance with a multi-scale enhancement strategy, thereby offering distinct advantages in multimodal semantic understanding and segmentation performance.

TMENet has made specific improvements to the existing architecture. Still, in the future, it can further construct a highly robust and highly generalized crack recognition system for real-world scenarios by introducing super-resolution automatic localization and text guidance on crack causes and types, and provide more accurate and effective technical support for road maintenance and decision-making.

VI. Conclusion

The detection of road cracks in complex scenarios is challenged by road backgrounds such as water stains, oil stains, road-printed signs, and small cracks. To better address these issues, a text-guided multiscale crack image segmentation fine-tuning network, TMENet, based on complex backgrounds, is proposed. Firstly, we designed the TCE Module to fine-tune the Text Transformer in the Clipseg model to enhance its analytical ability for describing specific cracks. Secondly, we developed the Ms-Mamba Module to capture and compensate for local features at different scales comprehensively. The Ms-Mamba Module improves the sensitivity and restoration ability of the model to local and multiscale details in segmenting crack images by integrating multi-scale composite convolution and residual connections. We have designed an LED module to enhance the crack edge detection capability in complex scenes. This module fully combines deep visual features and high-frequency edge information, and uses a refined feature adjustment mechanism to distinguish cracks from background noise. Finally, TMENet achieved the best results in comparative testing with seven state-of-the-art specialized crack segmentation networks on its self-built CampCrack800 dataset and three publicly available datasets. In addition, TMENet has achieved excellent results in actual road testing, proving that TMENet can provide better road crack detection methods for engineering and technical personnel.

This article proposes a novel network, TMENet, for detecting complex road cracks. TMENet fine-tunes the multiscale crack segmentation network based on complex backgrounds. TMENet has designed the Textual Crack Enhancer module, the Multiscale Mamba module, and the Laplacian edge detection module to enhance the crack edge detection capability in complex scenes. The results of ablation, comparison, and generalization experiments indicate that TMENet outperforms seven state-of-the-art segmentation networks. Real road testing has demonstrated the feasibility of using TMENet for road crack segmentation in complex scenarios. Our method has broad application prospects in the field of engineering.

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Ruoli Yang is currently pursuing the Ph.D. degree at the National University of Defense Technology, China. Her research interests include digital human technologies, deep learning, computer graphics, and image processing.

Jie Jiang is currently a Professor at the Key Laboratory of Big Data and Decision Making, Department of Systems Engineering, National University of Defense Technology. His research interests include artificial intelligence and deep learning, visualization and visual analytics, virtual reality, and intelligent interaction.

Lixin Zhan received the M.S. degree in computer technology from Nanchang University, China, in 2024. He is currently pursuing the Ph.D. degree with the National University of Defense Technology, China. His research interests include 3D computer vision and deep learning.

Baiyan Wan is currently pursuing the degree with the National University of Defense Technology, has a keen interest in sciences.

She has specialized in her academic focus on deep learning, graphics, and image processing. She is passionate about advancing AI capabilities through the exploration and development of algorithms and technologies.

Rui Qi is currently pursuing the master's degree in management science and engineering with the National University of Defense Technology. He is an Avid Academic Enthusiast. He has a strong passion for the development and advancement of technology, with a particular interests in computer vision and 3D reconstruction. He aspires to make contributions to the academic field.

Guoxiong Zhou received the B.Sc. degree from Hunan Agricultural University in 2002 and the M.Sc. and Ph.D. degrees from Central South University, China, in 2006 and 2010, respectively. He is currently a Professor with the Central South University of Forestry and Technology. His main research interests include forest fire prevention and robotics.

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